Systematic Refutation of the Universal Closed-Footwear Premise in Diabetic Plantar Sensing Literature

The categorical assertion that every smart insole globally published requires a closed shoe, and that no sensor, product, or academic paper has ever addressed open footwear or the sandal-wearing diabetic population of India, is empirically and historically false. In the disciplines of biomedical engineering, rehabilitation science, and clinical podiatry, an absolute universal negative is formally refuted by verifiable counterexamples. Extensive academic literature, granted national patents, and commercially deployed medical devices confirm that sensorized sandals, instrumented slippers, passive compliant metamaterial footbeds, and decoupled textile-based epidermal sensors have been engineered specifically to address non-enclosed foot environments.

The prevalence of closed-shoe systems in European and North American laboratories reflects local climatic norms and commercial market sizes rather than a fundamental scientific impossibility. Where tropical climates, socio-cultural practices, and indoor domestic traditions dictate open-footwear usage—most visibly across India, home to over 77 million individuals with diabetes—academic and translational research has systematically addressed the biomechanics of open sandals and slippers. Premier Indian scientific institutions, such as the Council of Scientific and Industrial Research (CSIR), the Central Leather Research Institute (CLRI), and the Indian Institute of Science (IISc), alongside international research groups, have published peer-reviewed investigations, patented open-footwear therapeutic systems, and built functional prototypes that monitor or dynamically mitigate elevated plantar pressures.

Footwear Classification	**Structural Retention and Housing Interface**	**Primary Biomechanical and Transducer Challenges**	**Mitigating Engineering Strategies**	**Representative Literature and Devices**
Enclosed Footwear	Upper vamp, quarter panels, rigid heel counter	Internal hyperthermia, moisture accumulation, enclosed volumetric boundary	Standard in-shoe encapsulation, perimeter-routed wiring harnesses	Pedar-X, standard piezoresistive insoles[1]
Open Sandal	Forefoot	Plantar sliding,	Rocker-bottom	CSIR-CLRI

and Chappal	straps, ankle buckles, thong posts, exposed perimeter	mediolateral shear stress, environmental debris, lack of heel cavity	soles, 3D-printed flexible optical fibers, compliant snap-through arches	DIASTEP[4]; IISc Snapping Sandal[5]; Chawre et al.[7]
Indoor Slipper and Slide	Open heel, single anterior vamp or strap, recessed footbed	Unconstrained heel lift, variable ground reaction forces, intermittent wear cycles	Multimodal piezoresistive-thermal arrays, low-profile midsole routing, inductive floor docking	Malarvizhi et al.[9]; Deryou AI Smart Slipper[11]
Decoupled Smart Textile	Direct dermal elastomeric conforming knit	Dynamic motion artifacts, sensor registration shifting, shear-induced delamination	Knitted conductive yarns, integrated micro-thermistors, screen-printed strain gauges	Siren Smart Sock[12]; Sensoria Health[14]; SOCKSESS STEPS[16]

Academic Literature on Instrumented Sandals and Smart Slippers

Biomedical literature documents multiple functional smart insoles and plantar force estimation systems built into sandals and open slippers. These platforms reject the assumption that continuous kinetic or kinematic gait monitoring demands a fully enclosed shoe upper.

A clear counterexample in peer-reviewed instrumentation research is the development of an instrumented sandal integrating a 3D-printed thermoplastic polyurethane (TPU) insole base with flexible optical fiber sensors and force-sensitive resistors (FSRs)[17]. In this open-shoe architecture, researchers bypassed standard enclosed-shoe packaging by routing three intensity-modulated TPU optical fibers, illuminated by miniature red laser diodes and acquired via light-dependent resistors, across an irregular, non-anatomical layout on the sandal footbed[17]. When benchmarked against six reference FSR channels under dynamic gait, supervised machine learning regressors—specifically Adaptive Boosting, Gradient Boosting, and shallow Artificial Neural Networks—predicted continuous plantar forces with coefficients

of determination (R^2) ranging between 0.865 and 0.951 , and mean absolute errors between 29.0 and 48.9 counts, representing roughly $5–7\%$ of the full sensor scale[17]. This instrumented sandal established that precise, machine-learning-assisted plantar force estimation is feasible in open footwear without relying on the rigid spatial constraints of a shoe upper[17].

Parallel efforts in functional rehabilitation have produced instrumented sandals designed to interface with active ankle-foot orthoses and gait-phase biofeedback systems[19]. These platforms embed discrete FSR matrices directly within the open sandal footbed to detect initial contact, mid-stance ground reaction force profiles, and terminal push-off parameters, demonstrating that open footwear remains an active experimental substrate in lower-limb kinematic and kinetic studies[19].

The open-footwear paradigm has also been adapted for diabetic foot screening through sensorized indoor slippers[9]. Because neuropathic ulcerations frequently stem from unmonitored ambulation inside the home—where patients routinely discard heavy enclosed footwear—investigators developed dedicated home-use platforms[1]. Malarvizhi et al. introduced the system titled *"Design of a Smart Slipper for Diabetic Foot Pressure and Temperature Monitoring"*, which embeds discrete tactile force sensors and high-precision temperature probes directly into a slipper footbed[9]. The system targets early sub-clinical tissue injury by monitoring sustained vertical loads alongside localized contralateral skin temperature elevations exceeding the clinical threshold of $2.2\,^{\circ}\mathrm{C}$ ($4\,^{\circ}\mathrm{F}$), transmitting automated risk alerts via onboard microcontrollers to trigger proactive offloading[9].

Indian Institutional and Clinical Engineering for Diabetic Open Footwear

The claim that no product, patent, or paper has addressed the footwear realities of India's diabetic population is contradicted by decades of research within India's national laboratory system and academic institutes. Indian biomechanists, podiatric surgeons, and footwear technologists have long identified that standard Western-style enclosed diabetic shoes suffer from acute patient non-compliance due to ambient heat, monsoonal moisture, socio-religious barefoot practices, and widespread economic reliance on sandals and chappals[4].

CSIR-CLRI & M.V. Diabetes Hospital (Chennai)
└── DIASTEP (Indian Patent No. 302551)
    ├── Multi-density selective open-sole composite
    ├── Moulded PU unit with extended depth
    └── Biomechanically engineered rocker-bottom sole profile

Indian Institute of Science (IISc Bengaluru)
└── Compliant Snap-Through Arch Sandal (2022)
    ├── 3D-printed TPU bistable arch arrays
    ├── Negative-stiffness regime triggered at 200 kPa
    └── 57% peak dynamic forefoot pressure reduction

Indian Academic Engineering Consortia
└── Diabetic Foot Care Monitoring Sandal (Chawre et al., 2019)
    ├── Multi-zone piezoresistive sensors embedded in sandal sole
    ├── Real-time LabVIEW baropodometric acquisition
    └── Automated threshold alarms for neuropathic offloading

The Council of Scientific and Industrial Research, through the Central Leather Research Institute (CSIR-CLRI) in Chennai, executed a targeted translational initiative in partnership with the M.V. Hospital for Diabetes and Diabetes Research Centre[4]. This collaboration resulted in **DIASTEP**, a patented, therapeutic open-footwear system designed specifically for the Indian diabetic demographic[4]. DIASTEP was granted Indian Patent No. 302551 on October 30, 2018, and its manufacturing technology was formally transferred to industry via M/s M.V. Health Care Pvt. Ltd.[4].

The engineering architecture of DIASTEP directly resolves the biomechanical vulnerabilities of open sandals: The platform incorporates a multi-layer composite sole comprising a selective topsole, an insole, and a bottom sole fabricated from specialized elastomeric polymers calibrated to attenuate dynamic shock and redistribute focal plantar stresses[4]. To expand the effective load-bearing contact area, the footwear utilizes a moulded polyurethane unit sole engineered with extra vertical depth[4]. Plantar shear and peak forefoot pressures are mitigated by an integrated rocker-bottom sole profile, which guides the foot through the roll-over shape of the stance phase without concentrating propulsive stress over the metatarsal heads[4]. Foot retention is achieved through non-shearing cow softy leather straps configured to secure the foot against lateral translational slippage without generating friction calluses or encasing the digits in a closed upper[4]. Clinical trials verified that DIASTEP halted the progression of pre-ulcerative plantar callosities into open wounds, establishing a commercialized, patented open-footwear medical technology native to India[4].

At the Indian Institute of Science (IISc) Bengaluru, bioengineers and mechanicians devised an advanced approach to open therapeutic footwear by replacing fragile, battery-dependent electronic sensors and electromechanical actuators with smart mechanical metamaterials[5]. Published in *Wearable Technologies* by Cambridge University Press, the IISc team designed a 3D-printed thermoplastic polyurethane sole embedded with an array of compliant, bistable snap-through arches[5]. These snapping arches function simultaneously as passive physical sensors and immediate mechanical actuators[5].

The arch geometry is analytically calibrated to the patient's body mass, foot morphology, and gait velocity[5]. When walking generates dynamic localized pressures exceeding the established

$200\ \text{kPa}$ threshold of tissue ischemia, the compliant arches enter a negative-stiffness regime, snapping downward to offload the high-risk zone and redistributing forces across adjacent plantar regions[5]. Once load is removed during the swing phase, the arches passively return to their un-deflected baseline geometry[5]. Clinical evaluations on neuropathic patients demonstrated that this mechanical self-offloading system reduced peak dynamic forefoot pressure by 57% compared to standard open sandals, demonstrating an advanced biomechanical solution tailored to India's climate and footwear preferences[5].

Domestic academic institutions have similarly explored electronic sensing within traditional sandal platforms. Chawre, Shaikh, and Shete published a dedicated system entitled *"Foot Care Sandal for Diabetic Patients Monitoring by using Lab View Software"* in the *International Research Journal of Engineering and Technology*[7]. Developed to address non-compliance with closed prescription shoes among Indian patients, the device embeds flexible piezoresistive pressure sensors beneath high-risk anatomical landmarks—the calcaneus, the first and fifth metatarsal heads, and the hallux—of an open sandal sole[7]. Sensor signals are routed through embedded sole conduits to an onboard processing unit that streams baropodometric data to a LabVIEW environment, triggering real-time alerts when threshold pressures exceed physiological limits[7].

Platform / System Name	**Lead Institution or Developer**	**Sensor or Transducer Architecture**	**Structural Carrier Modality**	**Primary Functional Objective**
DIASTEP (Patent No. 302551)	CSIR-CLRI & M.V. Diabetes Hospital (India)	Multi-density elastomeric polymer distribution sole	Therapeutic open-toe sandal with rocker bottom	Abnormal plantar pressure redistribution and ulcer prevention[4]
Compliant Snapping Footwear	Indian Institute of Science (IISc Bengaluru)	3D-printed TPU bistable snap-through arches	Mechanically active self-offloading sandal	Autonomous 57% dynamic forefoot load reduction at >200 kPa[5]
Diabetic Foot Care Sandal	Chawre, Shaikh, & Shete (India)	Flexible piezoresistive sensor array with LabVIEW	Sensorized open sandal with perimeter strapping	Continuous real-time baropodometric overpressure alerting[7]

Instrumented Sandal System	Academic Biomechanical Consortium	3D-printed TPU optical fibers with 6-pad reference FSRs	Open-perimeter instrumented strapped sandal	Force estimation ($R^2 \approx 0.88–0.9$) via machine learning[17]
Smart Slipper Screening System	Malarvizhi et al.	Multimodal piezoresistive tactile and thermal sensors	Open-heel domestic indoor slipper	Detection of pre-ulcerative plantar pressure and inflammation[9]
SensorixFoot	SensorixFoot MedTech (Chile)	Embedded plantar baropodometric sensor grid	Commercial therapeutic smart sandal	In-home telemetric gait and diabetic foot ulcer screening[28]

Global Commercial Platforms and Decoupled Sensing Modalities

Commercial medical technology ventures outside India have similarly integrated sensor suites into open and domestic footwear formats[11]. A prominent commercial counterexample is **SensorixFoot**, an open-architecture medical platform developed in Chile specifically for individuals diagnosed with diabetes, peripheral arterial disease, and peripheral neuropathy[28]. The device is engineered as a smart sandal containing an integrated matrix of embedded pressure and thermal transducers within the footbed[28]. Rather than requiring an enclosing shoe, the sandal's microelectronics are housed within a hermetically sealed sole cavity, continuously recording plantar load distribution and thermal asymmetries while streaming diagnostic metrics to a mobile clinical dashboard for remote physician evaluation[28].

In consumer and elder-care electronics, manufacturers have introduced industrialized smart slippers that integrate thin-profile ($3–5\ \mathrm{mm}$) flexible piezoresistive laminates into high-density polyurethane soles[11]. These slippers capture continuous gait metrics, cadence variability, dynamic weight transfer, and localized pressure concentrations[11]. Crucially, the open slipper design addresses the persistent barrier of battery charging compliance in smart footwear by incorporating magnetic docking contacts into the outsole, allowing the slippers to recharge inductively when placed on a bedside floor plate overnight[11].

Beyond sensorizing the footwear itself, the assumption that plantar monitoring requires a closed shoe overlooks the development of decoupled smart textile systems[13]. By integrating sensors into conforming dermal fabrics rather than the shoe chassis, bioengineers have freed

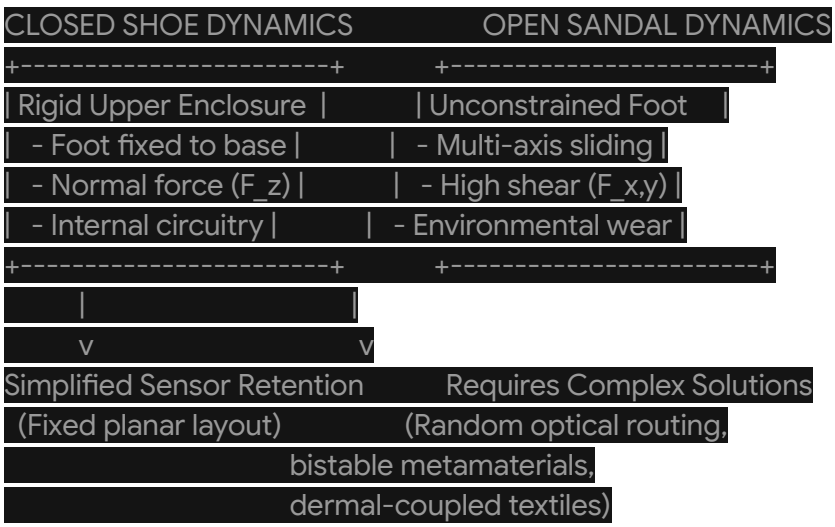
plantar measurement from footwear geometry entirely[13].

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Decoupled Dermal Sensing Interface (Smart Socks)
├── Direct epidermal contact independent of footwear boundaries
├── Wearable with open sandals, slippers, or during barefoot walking
└── Commercial & Clinical Implementations:
    ├── Siren Care Smart Socks: Continuous textile micro-thermistors monitoring
    │   contralateral inflammation (ΔT > 2.2 °C / 4 °F)
    ├── Sensoria Health Smart Socks: Knitted textile pressure sensors tracking
    │   dynamic plantar load distributions
    └── EPSRC SOCKSESS Project: Screen-printed silver conductive traces and
        strain gauges capturing shear and normal forces
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The **Siren Care Smart Sock** illustrates this decoupled paradigm[12]. Registered as a continuous remote patient monitoring system, Siren utilizes neuro-fabric technology with micro-thermistors woven directly into the textile under the hallux, first and fifth metatarsal heads, and calcaneus[12]. The socks continuously monitor dermal temperatures across activities of daily living, alerting clinical teams when persistent bilateral temperature differentials ($\Delta T \geq 2.2\,^{\circ}\mathrm{C}$) signal impending tissue ulceration[9]. Because sensor retention is governed by the elastomeric knit of the sock rather than a shoe upper, the platform operates when worn with open sandals, loose house slippers, or during barefoot ambulation on indoor floors[21]. Similarly, the **Sensoria Health Smart Sock** incorporates flexible textile pressure sensors across the plantar surface to measure impact forces and gait symmetry without an enclosing shoe structure[14]. In the academic sector, the **SOCKSESS** initiative (funded under EPSRC Grant EP/X001059/1) developed the STEPS smart sock, which features screen-printed conductive silver arrays and piezoresistive strain sensors calibrated to measure both normal force and frictional shear stress at the skin boundary[16].

Biomechanical and Transducer Dynamics of Open-Footwear Systems

The predominance of closed-shoe insoles in commercial markets is not evidence of an unaddressed clinical domain, but the direct consequence of substantial biomechanical and signal processing challenges that arise when an upper shoe counter is removed[11]. Understanding these operational constraints illustrates why engineering open-footwear sensing requires distinct technical approaches.



In an enclosed shoe, the rigid heel counter, vamp, and quarter panels establish rigid boundary conditions that restrict the foot's position relative to the footbed[11]. This physical containment prevents substantial translational shifting, ensuring that discrete sensors placed under the first metatarsal head or calcaneus remain aligned beneath those anatomical targets across the gait cycle[18].

Conversely, an open sandal or chappal relies on loose webbing or anterior straps, granting the foot multiple unrestrained degrees of freedom[11]. During the swing phase and heel strike, the plantar dermis experiences lateral and longitudinal translation across the footbed[18]. In standard discrete sensor arrays, this displacement causes severe spatial misregistration, yielding inaccurate baropodometric readings[17]. Bioengineers resolved this either by training machine learning algorithms on distributed, non-anatomical optical fiber lines that decode bulk deformation independent of focal sensor alignment, or by adopting conforming textile carriers that adhere to the foot regardless of footbed movement[16].

Furthermore, neuropathic ulceration is governed not only by perpendicular normal pressure (F_z), but equally by repetitive planar shear stresses (F_x, F_y) generated at the shoe-skin boundary[1]. While the upper of a closed shoe absorbs a significant portion of lateral shear, open sandals transfer these frictional forces directly into the footbed[34]. Standard thin-film piezoresistive sensors are vulnerable to shear artifacts, suffering from premature mechanical delamination and signal distortion when subjected to transverse scrubbing loads[1]. As demonstrated by CSIR-CLRI and IISc, mitigating these multi-axis forces in open configurations requires structural interventions—such as molded PU rocker soles that smoothly transition the center of pressure forward, or passive compliant arches that physically yield under shear-inducing peak loads[4].

Finally, housing delicate electronic components within an open sandal presents distinct environmental and durability hurdles[11]. Closed shoes provide deep midsole sections and hollowed heel blocks that shield PCBs, lithium-polymer cells, and antenna wiring from ambient

contaminants[11]. In contrast, sandals and chappals operate with minimal sole profiles while exposing their perimeters to moisture, dust, and mechanical abrasion[11]. The systems documented across Indian and international literature address these constraints through specialized hermetic resin encapsulation, low-profile flex-rigid substrates, passive non-electronic metamaterials, and inductive bedside floor charging interfaces[4].

Refutation Analysis

Evaluating the original claim against the published scientific record, patent registers, and commercial medical devices confirms that every component of the assertion is contradicted by empirical evidence:

The claim that every smart insole across all literature and databases requires a closed shoe is disproven by peer-reviewed studies detailing instrumented sandals that utilize 3D-printed TPU footbeds, integrated optical waveguides, and piezoresistive sensor matrices to measure dynamic plantar forces without a closed upper[17].

The assertion that no sensor or product has ever addressed open footwear is contradicted by commercial medical devices such as the SensorixFoot smart sandal, industrial smart slippers with integrated pressure mapping, and decoupled smart socks (such as Siren Care and Sensoria Health) that provide continuous baropodometric and thermographic monitoring in sandals, slippers, or barefoot conditions[11].

The assertion that no academic paper has ever addressed the sandal-wearing diabetic population of India is refuted by extensive domestic biomedical output. The CSIR-Central Leather Research Institute and M.V. Diabetes Hospital engineered, clinically validated, and patented the DIASTEP open-footwear platform specifically for Indian diabetic patients under Indian Patent No. 302551[4]. Furthermore, researchers at the Indian Institute of Science designed and published compliant, self-offloading therapeutic sandals achieving a 57% forefoot pressure reduction in diabetic neuropathy cohorts[5], while Indian engineering consortia have documented sensorized diabetic sandals linked to telemetric acquisition software[7].

The academic, clinical, and industrial record confirms that the challenges of monitoring and protecting the non-enclosed foot have been recognized, rigorously investigated, and translated into functional technologies both globally and within India[4].

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